

Arjun Samayam

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EDUCATION

M.S. Applied Artificial Intelligence

August 2024 - July 2025

Florida International University, Miami, United States

- Achieved a perfect **4.0/4.0** GPA in master's program, demonstrating strong performance

B.E. Electrical and Electronics Engineering

August 2020 - February 2024

Birla Institute of Technology and Science Pilani, Hyderabad, India

- Earned early graduation with **first**-division honors, a rare academic achievement

EXPERIENCE

Compliance Student Assistant - Office of The President

March 2025 - Present

FIU, Miami, United States

- Coordinated the development, implementation, and management of university policies and procedures, while managing record retention and document analysis to support policy updates, compliance, and accuracy
- Provided administrative support by coordinating tasks, managing communications, and ensuring timely completion of objectives, while handling additional responsibilities as required

Data Science Intern – R&D on AI and ML

August 2023 - December 2023

Genpact, Hyderabad, India

- Developed a customizable multi-tool agent using LangChain and Gen AI to build LLM-powered applications
- Proposed an innovative document processing feature for the Genpact AP helpdesk platform, adapting use cases like web search, OCR extraction, summarization, Q&A, semantic search, database retrieval, and image generation

COURSES

AI principles, Machine Learning, Generative AI, Deep Learning, Data Visualization, Big Data Analytics, AI for Robotics, Cloud Computing, Embedded AI, Cyber-Physical Systems and Security, Computer Architecture, VLSI Design, Smart Grid, Control Systems, Power Electronics, Principle of Economics, Disability Studies, International Relations

SKILLS

Artificial Intelligence, Langchain, Tableau, PowerBI, Machine Learning, Python, SQL, Prompt Engineering, Large Language Models, AWS, IoT, MATLAB, OpenCV, Image Processing, C Programming, AutoCAD, Verilog, LTSpice

CERTIFICATES

Microsoft Certified: Azure AI and Data Fundamentals, AI For Everyone, Google Generative AI Learning Path, Competing in Business AI-Powered Supply Chains, Mastercard Cybersecurity and Verizon Cloud Platform Virtual Experience Programs, Hacker Rank Python and Problem Solving, Postman Student Expert and API Programming

AWARDS

Hackathon Winner, Genpact

October 2023

- Emerged victorious in Skill Showcase Hackathon Challenge organized by the Genpact AI/ML Practice team, demonstrating strong collaboration and technical expertise

Academic Distinction, RBS

March 2018

- Earned the **Best Outgoing Student** award as the first to attain a perfect **10/10** score, recognized for outstanding performance in both academic and non-academic pursuits of schooling

COMPETITIONS

Educational Assessment Australia

September 2015

- Achieved top **25%** in English and Mathematics in the International Assessments for Indian Schools IAIS organized by the University of New South Wales UNSW, Sydney, Australia

PROJECTS

Unmasking AI Generated Images: Deepfake Detection using CNNs| Miami

Deep Learning Research and Implementation

October 2024 - December 2024

- Constructed a Convolutional Neural Network based framework for detecting deepfakes and AI-generated images
- Performed batch-wise analysis on test data, revealing performance variability and insights for future model refinement and generalization across diverse datasets

Advanced Bike Sharing Performance Analytics| Miami

Data Analysis and Visualization

October 2024 - December 2024

- Created interactive Tableau dashboards to assess bike-sharing trends. Performed data cleaning, handled missing values, and formulated calculated measures to evaluate rental growth, user proportions, and weather impacts
- Designed dynamic visualizations to showcase total rentals, external factors, and seasonal trends, enabling insights

Machine Learning for Edge IIoT Intrusion Detection| Miami

Cybersecurity and AI principles

August 2024 - October 2024

- Developed and evaluated ML models for intrusion detection in Edge and Industrial IoT networks, securing near-perfect accuracy with Random Forest and AdaBoost classifiers
- Executed comparative analysis of multiple algorithms including Decision Trees, Random Forest, SVM, and KNN, testing effectiveness based on accuracy, sensitivity, precision, and F1-score metrics

Smart Weather Monitoring System| Miami

IoT and Embedded Systems

August 2024 - October 2024

- Devised a Raspberry Pi-based weather monitoring system for real-time environmental data collection, including temperature, humidity, light intensity, and motion detection, and image capture using camera module
- Integrated multiple sensors for enhanced functionality and accuracy. Enabled real-time data transmission to Adafruit IO, featuring an interactive LCD and LED matrix display for remote monitoring

Data Analysis and Classification| Hyderabad

Cyber Physical Systems and Security

March 2023 - May 2023

- Built Python functions with Scapy to analyze CIP packets, extract unique SCADA tags, and identified one-to-one mappings between tags and IP addresses. Calculated time gaps for each packet class and visualized results

RISC-V Pipelined Processor| Hyderabad

Computer Architecture Digital Design

February 2023 - April 2023

- Implemented single-cycle and pipelined RISC-V processor with four stages: Fetch, Decode, Execute, Write Back
- Operated R-type, I-type, B-type, and J-type instructions. Implemented Forwarding to resolve data hazards

Power System Analysis and Control| Hyderabad

Enhancement and Stability Optimization

January 2023 - April 2023

- Created and analyzed a three-bus power system using stabilizer design, fault impact and voltage optimization

Obstacle Avoidance| Hyderabad

Robotics and Control Systems

October 2022 - December 2022

- Designed an automated navigation system using MATLAB for real-time obstacle detection and avoidance

Secure Cloud Deployment| Hyderabad

Cloud Computing and Security

September 2022 - November 2022

- Devised and deployed a secure cloud-based web application and database persistence system on AWS, utilizing a bastion host and virtual private cloud architecture to ensure safe server management and enhanced security

Myth buster: Stereotypes about disabilities| Hyderabad

Social Inclusion and Awareness

September 2021 - November 2021

- Challenged disability stereotypes and improved campus accessibility to encourage an inclusive environment